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Paleomagnetic evidence for rotation of the Iberian Peninsula and the external Betic Cordillera, Southern Spain

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ABSTRACT

Paleomagnetic investigations of Jurassic, Cretaceous and Tertiary sites in the external Betic Cordillera indicate that, while the observed remanence vectors have inclinations which are conformable with the expected inclinations for stable Iberia, declinations deviate significantly. These results indicate that: (1) the Mesozoic limestones of the Penibetic have been rotated clockwise with respect to stable Iberia and, (2) there is a differential rotation between the Cretaceous sites (Dec. = 61° ± 11°) and the Jurassic sites (Dec. = 38° ± 8°). The differential rotation of 23° between the late Jurassic and late Cretaceous is consistent with the Cretaceous anticlockwise rotation of the Iberian Peninsula with respect to Eurasia. It follows that during this time period this part of the Betic mobile belt was part of the continental margin of the Iberian Peninsula. A study which was confined to a single unbroken stratigraphic section west of Ronda indicates that most of the anticlockwise rotation (20°) probably occurred between the Cenomanian and the Campanian. The anticlockwise rotation of these rocks must have been followed by a clockwise rotation of at least 60° to account for the observed clockwise declination anomalies.

1. Introduction

The Betic Cordillera of southern Spain forms the northern limb of the tightly curved Alpine orogenic system that lies at the western end of the Mediterranean collisional system. The external zones of this mountain chain represent the continental margin of the Iberian Peninsula that was detached from its basement as a result of Miocene convergence between Africa and Eurasia. An understanding of the geometry and kinematic development of this mountain chain therefore has fundamental implications both for the Alpine orogenic development of the Gibraltar arc and for the pre-Miocene kinematics of the Iberian Peninsula.

Since the Mesozoic breakup of Pangea perhaps the most significant relative movement between Iberia and Europe occurred in the middle to late Cretaceous as a consequence of the opening of the North Atlantic, when Iberia rotated anticlockwise away from France and opened the Bay of Biscay. Evidence for rotation of the Iberian Peninsula is abundant. Geological and geophysical data obtained from the Bay of Biscay indicate that the rotation amounted to approximately 30° [1]. Results of drilling at DSDP sites 401, 402 and 403 [2] establish that the end of the late Jurassic to early Cretaceous rifting phase and the commencement of the opening of the Bay occurred after anomaly M-0 (late Aptian–early Albian). The youngest magnetic anomalies, which are present in the central part of the bay, are probably anomaly numbers 33 and 34 (Campanian–Maastrichtian) [3,4]. However, because of the early Tertiary collisional event, which lead to the welding of Iberia to Europe and to the blurring of the magnetic anomalies in the Bay of Biscay, there is still some uncertainty and debate as to when the extension associated with the opening of the bay actually ceased [5].

The paleomagnetic data obtained from the Iberian Peninsula is in general agreement with the geological and geophysical data from the Bay of Biscay [6–14], and the paleomagnetic evidence

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gives a rotation that may have been 35°–41°. Galdeano et al. [14] have studied sediments and volcanics west of Lisbon and suggested that 27° of the total rotation occurred very rapidly between 125–110 Ma and 14° occurred at some later time.

Previous paleomagnetic studies in the external Betic Cordillera established the existence of lithologies that were suitable for paleomagnetic studies. They also demonstrated that there were significant rotations in both the clockwise and the anticlockwise directions [12,15–17]. The most recent of these studies [16,17] showed that the Jurassic volcanics of the Subbetic Zone northwest of Granada were rotated consistently clockwise by approximately 50°. The other studies [18–21] were primarily magnetostratigraphic.

In this paper paleomagnetic data record large clockwise rotations about vertical axes in the western external Betic Cordillera. The results have implications for the orogenic development of the Betic Cordillera and its participation in the Mesozoic rotation of the Iberian Peninsula.

2. Geological setting

The present geological configuration of the Betic Cordillera can be broadly divided into an internal and an external realm [22] based on paleogeographic and structural criteria (Fig. 1). The Internal Zone consists of those lithostratigraphic units which lie on the concave side of the orogenic belt or are the farthest from the Iberian and African continental margins. They are composed of an extensive exposure of Triassic and sometimes younger cover sequences resting on a Precambrian to Paleozoic basement. These rocks are usually grouped into three syn- to post-metamorphic tectonic complexes which were affected by varying degrees of metamorphism and pene-

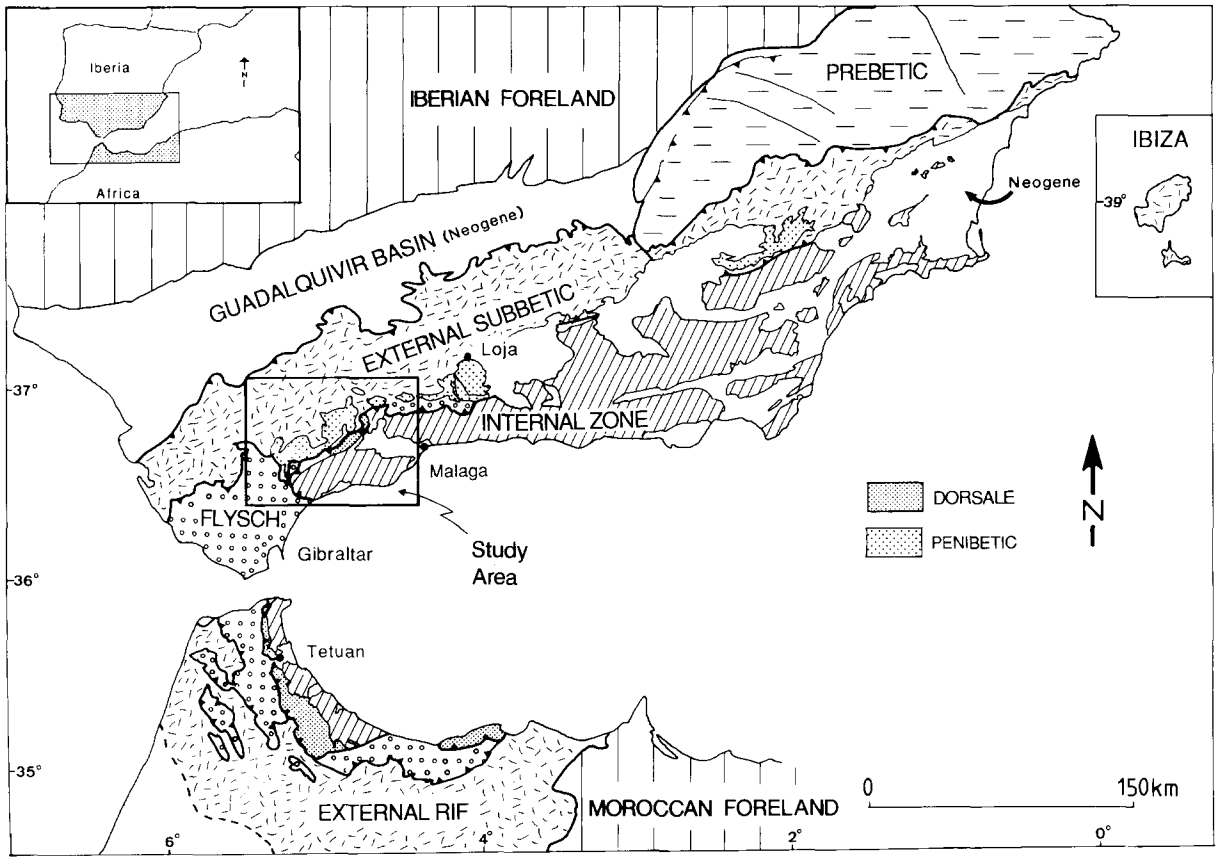


Fig. 1. Simplified geological map of the Betic/Rif orogenic belt.

trative deformation. Deformation in the Internal Zone may have started as early as late Cretaceous time, and was largely completed by mid-Miocene time.

The External Zones are generally thought to represent the passive continental margins of Iberia and Africa which were deformed during the Alpine orogeny. In contrast to the Internal Zones, these rocks are exclusively of Mesozoic and Tertiary age. In the External Zones the Paleozoic basement is most likely a continuation of the Hercynian of the Iberian/Moroccan Meseta that is detached from its deformed Mesozoic cover sequence along an Upper Triassic, salt-rich, Keuper horizon. In addition, the sediments of External Zones were generally unaffected by Alpine metamorphism. In Spain the External Zone is generally divided into a number of units which include the Prebetic and the Subbetic Zone (Fig. 1).

Prebetic Zone: Due south of the Spanish Meseta, or foreland, and in the northeastern sector of the Betic/Rif orogenic belt is the Prebetic Zone. This zone contains relatively thin Mesozoic to Tertiary neritic and continentally derived sediments in its more proximal reaches and platform facies carbonates with minor siliclastic sediments in its more distal parts. The crust is parautochthonous and was affected in some areas only by a post-Oligocene compression. The character of the deformation in this area has been compared to that in the Jura mountains of eastern Switzerland [23].

Subbetic Zone: The Subbetic Zone represents a basinal sequence which is the continuation of the Prebetic Zone. This allochthonous sequence was folded and thrust north to northwestward during the later phases of the Alpine orogeny. It is composed of Mesozoic to Tertiary pelagic to deep-water marine sediments. The Mesozoic paleogeographic evolution of the Subbetic Zone has been summarized by Garcia Hernandez et al. [24]. Due to the episode of extension and rifting which began in the middle to late Jurassic and resulted in the breakup of the Liassic carbonate platform, the details of the Subbetic stratigraphy vary along the strike of the zone and across the zone from the more internal to the more external Subbetic realms.

The oldest sediments that outcrop in the Subbetic Zone are Triassic in age. These sediments are comparable to the Germanic Triassic sequences described from the northern Alps and consist of Buntsandstein, Muschelkalk and Keuper facies. This Triassic stratigraphy suggests that the External Zone corresponds to a paleogeographic domain that is distinct from the Internal Zone.

In early Jurassic time there was a marine transgression. Extensive shallow-water platforms were created upon which carbonate rocks were deposited. These carbonates are commonly at least partially dolomitized. At the beginning of the mid-Jurassic this region began to experience differential subsidence related to the extension associated with the breakup of Pangea. The morphology of the continental margin in the late Jurassic was characterized by a series of submarine swells and troughs. In the Subbetic realm pelagic conditions were established. As has been observed elsewhere along the Tethys margin [25], a condensed red to white nodular limestone sequence rich in ammonites (the *Ammonitico Rosso*) was deposited on the submarine highs. Sedimentation rates for these Upper Jurassic condensed limestone sequences as determined in the Sierra de Teba [26] are of the order of 3.5–5.5 mm/10^3 yr. These sediments have been the subject of many recent paleomagnetic and magnetostratigraphic studies both in Spain [18,19,21] and elsewhere in the Mediterranean [11,27,28,29,30] because they carry a strong stable magnetization.

In some areas these platforms were not subject to currents and as a result are composed of grey, well-bedded micritic and sparry limestones. Off the edges of these submarine swells oolitic limestones with benthic foraminifera were deposited and redeposited. In the deepest basins where sedimentation was occurring below the carbonate compensation depth (CCD) radiolarian cherts and calcareous turbidites were deposited. The middle to late Jurassic extension ultimately led to basaltic volcanism in these deeper troughs [31].

During the Cretaceous, sedimentation gradually became uniform over the entire External Zone. The lower Cretaceous buff to yellow marls and marly limestone sequences are of variable thickness and occasionally incorporate breccias, sedimentary slumps or detrital turbidites. Sometimes the lower Cretaceous is completely absent

from the sequence and is represented only by hardgrounds or submarine karsts [32].

The upper Cretaceous to Eocene deposits of the Subbetic Zone are a marl to marly limestone sequence known as the *Capas Rojas*. These sediments are correlative with the *Couches Rouges* in France and the *Scaglia Rossa* in northern Italy. Like the *Ammonitico Rosso* they carry a strong, stable magnetization, and have been used for paleomagnetic and magnetostratigraphic studies elsewhere in the Mediterranean [11,27,28,33]. They are characteristic of a deep-water pelagic sedimentary environment which has a fine detrital input.

The Internal/External boundary west of Loja is delineated by the most internal of the Subbetic paleogeographic units (Fig. 1). This unit is often referred to as the Penibetics after Blumenthal [22]. The Penibetics are distinguished stratigraphically and therefore paleogeographically from the remainder of the Subbetic Zone. Throughout the Jurassic rifting phase this unit never subsided below the CCD but remained a relatively shallow platform. In the late Jurassic, while oolitic limestones were being deposited in the Penibetics radiolarian cherts were being deposited both in the medial Subbetic Zone to the north and to the west of the Penibetics, and in the Internal Zone to the south and to the east of the Penibetics. Finally, in the stratigraphic succession of the Penibetics, the lower Cretaceous is often very thin or entirely missing [32,34]. It is therefore not uncommon to find a karstified late Jurassic sequence lying directly below the late Cretaceous.

Although there is a great deal of literature describing the stratigraphy of this part of the Betic Cordilleras, little detail is known about its structure. For a summary of both the theses and the published literature on the Penibetics, see Martin-Algarra [34] and Azema et al. [35]. Morphologically the Penibetics is a region divided into a series of basins and ranges. The late Jurassic platform carbonates form the ranges while the late Cretaceous *Capas Rojas* formation outcrops in the basins.

The topographic relationship between the late Jurassic and the late Cretaceous limestones suggests that the stratigraphy has been inverted. However, the contacts between these two units may be normal, inverted, conformable and stratigraphic, or tectonic. In some areas, the stratigraphic succession has remained gently dipping and homoclinal despite the significant amount of regional shortening that has occurred as a consequence of the Alpine orogenic event. In these regions normal, upward younging stratigraphic contacts are observed. In other locations, where the local deformation has been more severe, the stratigraphic sequence may still be gently dipping but completely overturned. These inverted, downward facing, stratigraphic contacts occur most commonly within the inverted limb of a footwall syncline which has formed under an overriding thrust sheet.

Although conformable stratigraphic contacts between the Jurassic and Cretaceous rocks are not uncommon in the Penibetics, most of the contacts are tectonic. In some instances the late Cretaceous has been faulted down into the basins along high-angle normal faults. In the majority of cases, however, the late Cretaceous lies beneath overthrust Jurassic thrust sheets.

The amount of penetrative strain suffered by the Mesozoic carbonate sequences of the Penibetics varies according to rock type. In spite of the late Alpine deformational episode the Jurassic limestones show little penetrative deformation. Most of the deformation in these carbonates has been in the form of translation and rotation of tectonically separated fault blocks. In the Penibetics these fault blocks appear as a series of homoclinally dipping units resting on the tops of the ranges.

In contrast to the competent Jurassic limestones, the late Cretaceous marly limestones of the *Capas Rojas* formation are often highly deformed on the outcrop scale. This formation, because of its incompetent marly lithology, appears to have taken up much of the shortening. Isoclinal folds, kink folds and cleavage are common.

Along the strike of the Penibetics the trend of the fold axes in the Mesozoic carbonate sequences changes, paralleling the change in strike of the arc. In the northeast section of the Penibetics, near the town of Antequera, the axes trend approximately E–W while further southwest, near the town of Ronda, they trend NE–SW. South of Ronda, in the N–S trending section of the Cordillera, there are only scattered outcrops

of the Subbetic carbonates. These outcrops, like the flysch nappes that are dominant in this region, show N–S trending fold axes.

3. Paleomagnetic and rock-magnetic methods

In this study 40 sites were sampled in the External Zone west of Loja. Ten to fourteen cores per site were collected from six to fourteen beds spread over an area dependent on the specific rock type. Samples were collected using either a portable gas drill or as oriented blocks. The samples were oriented using a Brunton compass and an orienting platform. The pelagic limestones sampled in this study were weakly magnetized and so did not significantly affect these orienting procedures. Standard cylindrical cores of 2.5 cm in diameter and 10 cm in length were later drilled and cut in the laboratory into 2.25 cm long samples.

In the laboratory the samples were analyzed to isolate the stable characteristic component of magnetization (ChRM), to establish the magnetic properties of the sediments and to characterize the minerals responsible for the magnetization of the rock.

To isolate the primary component of magnetization, stepwise thermal and alternating field demagnetization procedures were used. During the thermal demagnetization procedure each sample was demagnetized with a minimum of ten steps based in a scheme designed after a detailed pilot demagnetization study. For this procedure a Schonstedt thermal demagnetizer with a residual field of approximately 10 nT was used. Measurement of the NRM (natural remanent magnetization) of the samples was made on a three-axis ScT cryogenic magnetometer [36]. The data obtained from the demagnetization experiments were plotted on Zijderveld vector diagrams [37,38]. Paleomagnetic directions could then be determined by using techniques such as linear regression [39].

On a set of representative samples, bulk susceptibility was measured between each thermal demagnetization step on a KLY-1 susceptibility bridge. These bulk susceptibility studies were used to monitor possible changes in magnetic mineralogy that can occur as a result of heating and dehydration.

Isothermal remanent magnetization (IRM) experiments performed on a representative number of samples (≈ 45) were used to characterize the properties of the magnetic carriers and thereby to determine the magnetic mineralogy of the sediments. Acquisition experiments were conducted at room temperature using an electromagnet with a maximum field strength of 1.1 T. As a final step, to examine the contribution to the magnetization from the low-coercivity minerals, the samples were placed in a field of 0.05 T along an axis normal to the previously defined axis. Subsequent thermal demagnetization of the two-component IRM gave the thermal decay history of both the high and the low coercivity components of magnetization.

In addition to the above-described procedures, the anisotropy of magnetic susceptibility (AMS) was measured on a select group of sites with a Digico anisotropy delineator spinner magnetometer [40]. A summary of AMS theory and measuring procedures is given by Hrouda, [41] and Borradaile [42]. The purpose of these experiments was to ascertain whether the sediments had acquired a magnetic fabric. Fabric development in sediments is often the result of bulk strains imposed on the sediment. This deformation frequently occurs either during compaction of the sedimentary layer, resulting in a sedimentary fabric, or during a tectonic event such as the the Alpine orogeny, resulting in a tectonic fabric which may distort the NRM directions.

4. Paleomagnetic and rock-magnetic results

The results of this work are presented in Table 1a and b and in Fig. 2. Of these forty sites, seven were cored in the lower–middle Jurassic pelagic limestones, twenty in the upper Jurassic pelagic series, eleven in the late Cretaceous, *Capas Rojas* formation and two in the late Miocene (Tortonian) limestones of the Ronda Basin.

All seven sites in the early Jurassic limestones, four sites from the late Jurassic series, and two sites sampled in the post-Alpine Miocene deposits, proved to be of no use for this paleomagnetic study. The early Jurassic limestones were either remagnetized in a present-day magnetic field (RL01, FJ03), too weakly magnetized (FJ01), or had a large within-site dispersion ($\alpha_{95} > 15°$)

(DJ04, DJ05). Site FJ02 and Jurassic sites AJ04, GJ01 and RJ13 gave valid directions but could not be adequately corrected for local structures. Finally, late Jurassic sites RJ09 and RJ11 had a large within-site dispersion, like sites DJ04 and DJ05.

The Miocene sites were sampled in the carbonate rich Las Mesas formation of the Ronda Basin. All the other rock types in the basin were either coarse-grained sandstones and conglomerates or fine-grained arenites too poorly lithified to sample for this paleomagnetic study. The intensity of the NRM of the algal carbonates of the Las Mesas formation was extremely low ($\leqslant 5 \times 10^{-5}$ A/m). Those samples that had enough intensity to be demagnetized usually had a large viscous component, such that the magnetization was unstable.

4.1. Late Jurassic pelagic limestones

The late Jurassic platform carbonates of the External Zone vary widely in lithology. Red to white micrites, grey sparites and white oolitic limestones are present within this zone. Although the rock-magnetic characteristics of these various limestones show significant differences the stable

TABLE 1

Remanent magnetization parameters for (a) sites used in this study and (b) sites not used. N = total number of sites. Columns give the site number and name, number of samples used per site (n), age of the outcrop (EJ = early Jurassic, LJ = late Jurassic, LC = late Cretaceous), declination (DEC), inclination (INC), k and the circle of confidence α_{95} both before and after tectonic tilt correction, and the average dip/dir direction of the bedding

a

N= 27					BEFORE DIP CORRECTION				AFTER DIP CORRECTION				Av. Bedding
no.	SITE	n	Age	Lat °N/Long °W	DEC	INC	α95	k	DEC	INC	α95	k	dip°/dir°
1	AJ01	7	LJ	37.0/ 4.5	49.3	41.4	8.9	47.2	38.4	40.2	7.9	59.4	15.0/323
2	AJ02	7	LJ	37.0/ 4.5	19.7	34.1	7.7	63.0	23.7	33.4	7.6	63.7	6.0/105
3	AJ03	11	LJ	37.0/ 4.5	34.1	9.8	4.8	92.7	43.5	60.6	4.8	92.7	50.0/205
4	AJ05	12	LJ	37.0/ 4.7	20.7	45.9	4.5	94.9	42.5	38.0	4.5	94.4	25.0/104
5	CC01	4	LC	36.4/5.3	46.1	30.6	43.3	5.5	45.8	46.7	12.2	58.1	46.0/318
6	CC02	9	LC	36.4/5.3	39.3	61.7	4.1	155.3	38.5	51.2	4.1	155.8	20.0/045
7	CJ01	9	LJ	36.4/5.3	64.0	64.9	4.6	125.5	67.2	53.8	4.6	125.5	15.0/080
8	CJ02	9	LJ	36.4/5.3	69.4	61.7	3.7	171.3	74.0	54.1	3.7	195.6	10.0/115
9	RC01	11	LC	38.1/ 5.1	86.6	25.1	6.4	53.5	64.9	41.6	4.8	93.3	35.0/321
10	RC02	6	LC	36.7/5.25	21.7	224.3	21.9	10.3	31.2	38.1	5.6	139.7	48.0/128
11	RC03	9	LC	37.0/ 4.9	56.8	27..5	6.5	64.1	73.1	20.7	5.9	77.8	34.0/145
12	RC04	9	LC	36.8/5.25	102.4	64.0	16.5	10.7	65.3	30.3	5.6	84.1	40.0/034
13	RC05	11	LC	36.8/ 5.3	66.9	67.5	5.2	79.1	69.0	29.7	5.2	79.1	38.0/070
14	RC06	12	LC	36.8/ 5.3	105.1	28.8	3.7	135.3	80.6	34.1	3.7	141.0	36.0/355
15	RC07	10	LC	36.8/ 4.9	37.0	59.8	6.2	62.7	53.3	36.7	6.5	56.5	27.0/076
16	RC08	11	LC	36.9/ 5.0	48.3	63.2	5.4	71.5	48.9	48.2	5.4	71.5	20.0/048
17	RC09	10	LC	36.7/ 5.2	56.9	49.6	5.8	71.5	55.2	41.5	5.1	89.9	11.0/120
18	RJ01	7	LJ	36.7/ 5.3	57.1	33.0	6.3	93.5	39.9	48.9	6.5	86.3	25.0/275
19	RJ02	7	LJ	36.8/ 5.1	49.0	49.1	7.0	75.5	20.7	47.4	7.0	75.5	25.0/310
20	RJ03	8	LJ	37.0/ 4.9	37.0	26.4	5.9	76.5	43.8	49.7	5.7	82.0	24.0/199
21	RJ04	10	LJ	36.8/ 5.4	39.1	49.7	4.6	128.7	77.0	34.9	4.2	151.8	42.0/130
22	RJ05	8	LJ	36.8/5.3	101.6	46.7	6.8	67.4	65.7	44.0	6.5	74.6	32.0/358
23	RJ06	8	LJ	36.9/5.1	40.7	69.3	3.2	308.5	58.3	45.5	3.2	308.5	26.0/074
24	RJ07	10	LJ	36.9/5.0	50.4	52.7	10.6	21.7	25.3	57.7	10.6	21.7	20.0/300
25	RJ08	8	LJ	36.8/5.4	278.6	37.2	6.6	65.6	6.2	41.6	7.7	52.9	76.0/055
26	RJ10	6	LJ	36.7/5.3	9.3	20.4	9.8	47.5	29.4	44.2	8.2	68.4	40.0/145
27	RJ12	10	LJ	36.7/5.1	12.7	32.1	9.3	28.0	33.7	37.7	9.1	29.1	30.0/125

b

N= 13					BEFORE DIP CORRECTION				AFTER DIP CORRECTION					Bedding
no.	SITE	n	Age	Lat °N/Long °W	DEC	INC	α95	k	DEC	INC	α95	k	comments	Av. dip°
28	AJ04	11	LJ	37.0/4.6	47.6	67.9	5.2	71.5	102.3	57.6	5.5	63.0	poor structural correction	27.0
29	DJ04	4	EJ	36.6/5.3	51.8	43.2	17.2	29.7	66.6	31.9	17.2	29.4	scatter too large	18.0
30	DJ05	7	EJ	36.6/5.3									low intensity	17.0
31	FJ01		EJ	36.5/									low intensity	44.0
32	FJ02	10	EJ	36.5/	199.0	30.4	7.2	46.6	213.5	48.2	7.1	47.3	poor structural correction	25.0
33	FJ03	7	EJ	36.5/	350.6	56.1	26.6	6.1	55.8	80.4	26.0	6.3	present magnetic field	30.0
34	GJ01	9	EJ	36.1/5.4	242.0	4.7		26.3	155.0	14.8	14.3	13.9	bad structural correction	51.0
35	RJ09	8	LJ	36.6/5.4									low intensity	60.0
36	RJ11	7	LJ	36.7/5.1	15.2	43.2	22.3	8.3	5.9	42.9	22.3	8.3	present magnetic field	10.0
37	RJ13	9	LJ	36.7/5.1	49.3	44.8	5.6	85.3	96.5	49.3	5.3	66.9	poor structural correction	?
38	RL01	6	EJ	36.9/5.4	352.8	56.5	2.3	885.2	348.6	34.9	2.2	909.9	present magnetic field	22.0
39	RM01	0	TM	36.9/5.2									low intensity	26.0
40	RM02	0	TM	36.8/5.1									low intensity	00

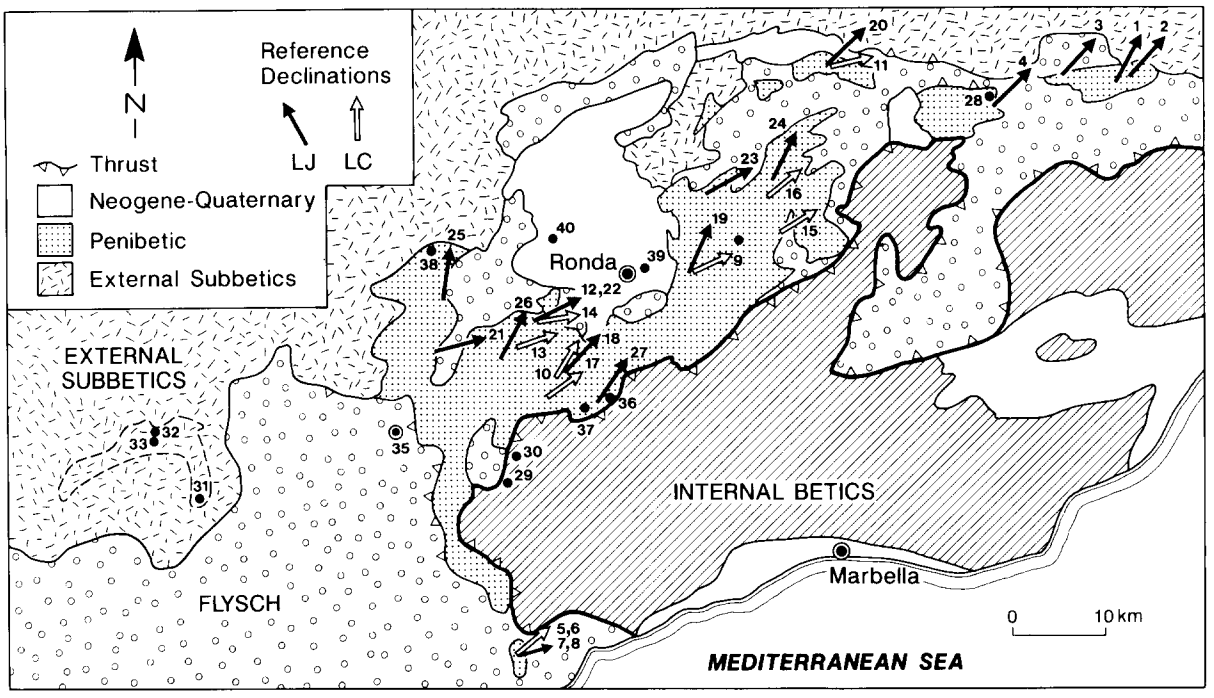


Fig. 2. Paleodeclination map of site means obtained from the Penibetic Zone west of Loja. *LC* = late Cretaceous; *LJ* = late Jurassic. Reference declinations calculated from [45]. Numbers correspond to sites as indicated in Table 1.

ChRM is carried by magnetite in most cases and the directions obtained are statistically well grouped ($\alpha_{95} < 10°$).

The intensity of NRM of the Jurassic limestones is bimodal. On average, the red limestones have a greater intensity of magnetization then the white or grey limestones. The direction of the NRM before bedding correction groups around the direction of the present-day magnetic field in southern Spain. This field is inclined approximately 53° towards the north.

The behaviour of the red nodular limestones during IRM acquisition and subsequent thermal demagnetization experiments (Fig. 3a and b) suggests that the rock-magnetic characteristics of the late Jurassic red limestones of the Penibetics are similar to those described in the studies of the *Ammonitico Rosso* from around the Mediterranean [16,18,43,44]. The experiments show that more than one magnetic mineral is present in these samples. A low-coercivity component, which is nearly saturated in fields of 0.1 T and is almost always demagnetized below 580°C, is most likely titanium-poor magnetite. A high-coercivity component, which does not saturate in the maximum applied field of 1.0 T and is only finally demagnetized at temperatures above 600°C, is probably hematite.

The IRM acquisition curves of some grey-white upper Jurassic limestones show a large low-coercivity fraction, then climb only slowly but steadily (Fig. 4a). The thermal demagnetization of the two-component IRM shows mainly decay of the low-coercivity fraaction up to 580°C, above which a small high-coercivity fraction remains until about 650°C (Fig. 4b). These grey-white limestones contain mainly magnetite, with a minor amount of hematite. Other grey-white limestones have IRM acquisition characteristics that still climb steeply in the maximum field of 1.1 T (Fig. 5a). During thermal demagnetization, the rapid destruction of most of the hard component below 100°C shows that the magnetization is carried by goethite (Fig. 5b). Further thermal demagnetization of the hard component of IRM shows the presence of magnetite and hematite as in the other lithologies. Small amounts of maghemite and sulphides may also be present in some lithologies, as indicated by a small dip in the unblocking temperature spectrum curves between

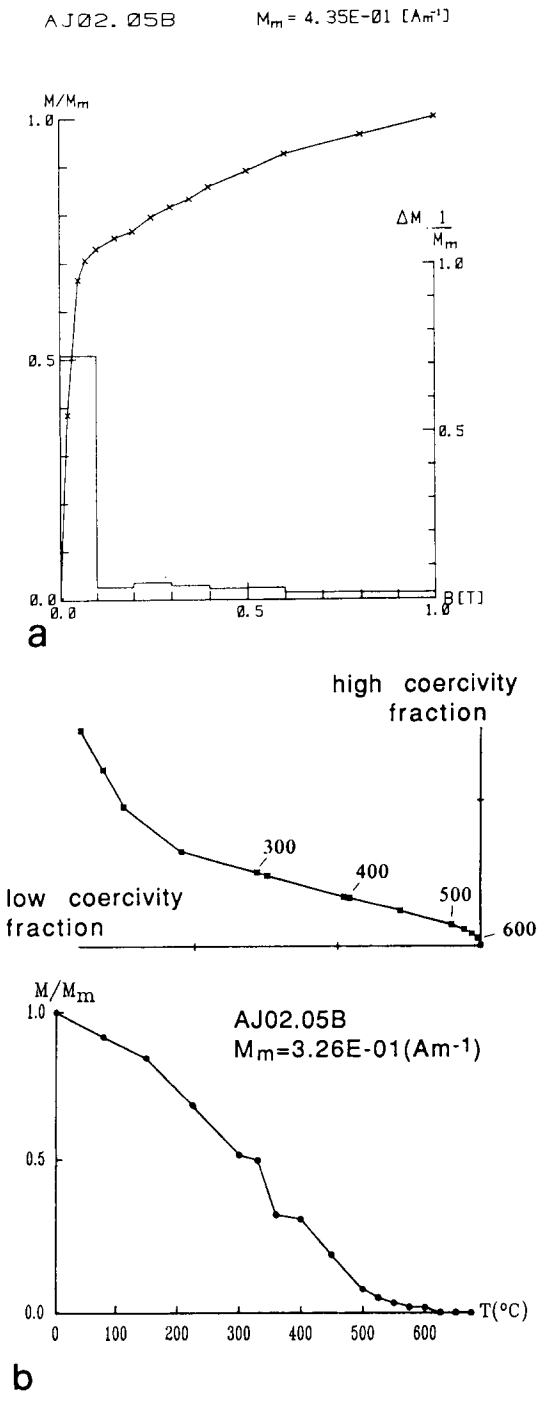


Fig. 3. (a) IRM acquisition curve and (b) vector diagram showing the thermal demagnetization behaviour of a two-component IRM for a sample of red nodular limestone from site AJ02.

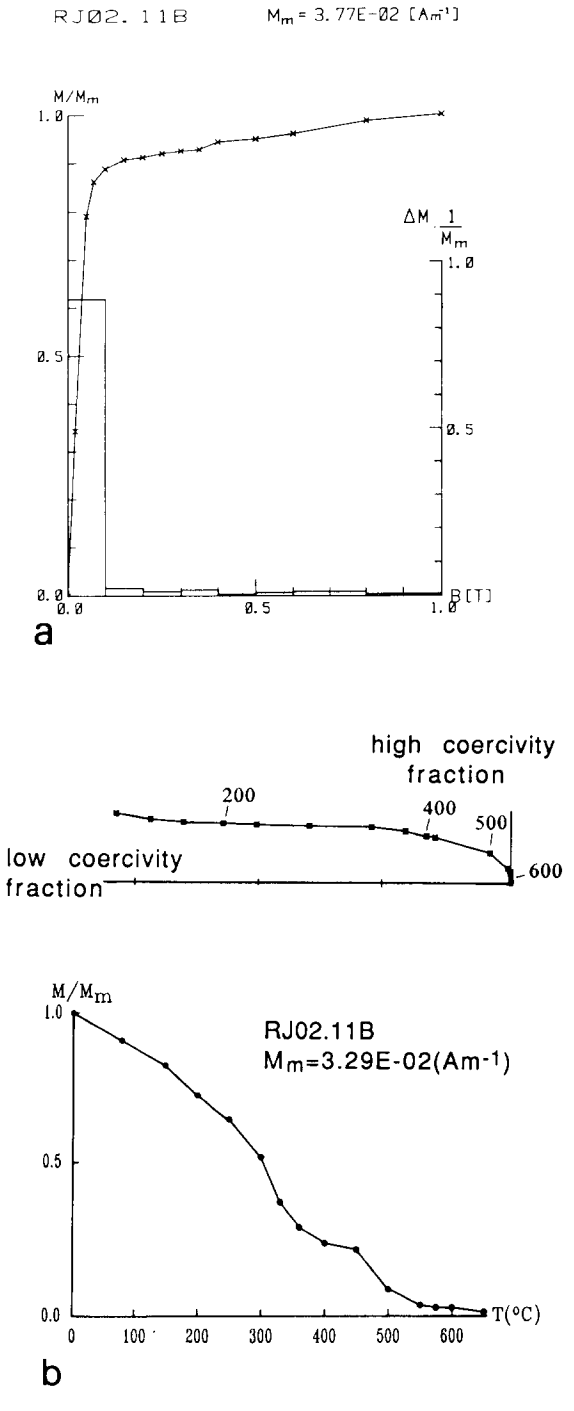


Fig. 4. (a) IRM acquisition curve and (b) vector diagram showing the thermal demagnetization behaviour of a two-component IRM for a sample cored from the grey-white pelagic limestones at site RJ02.

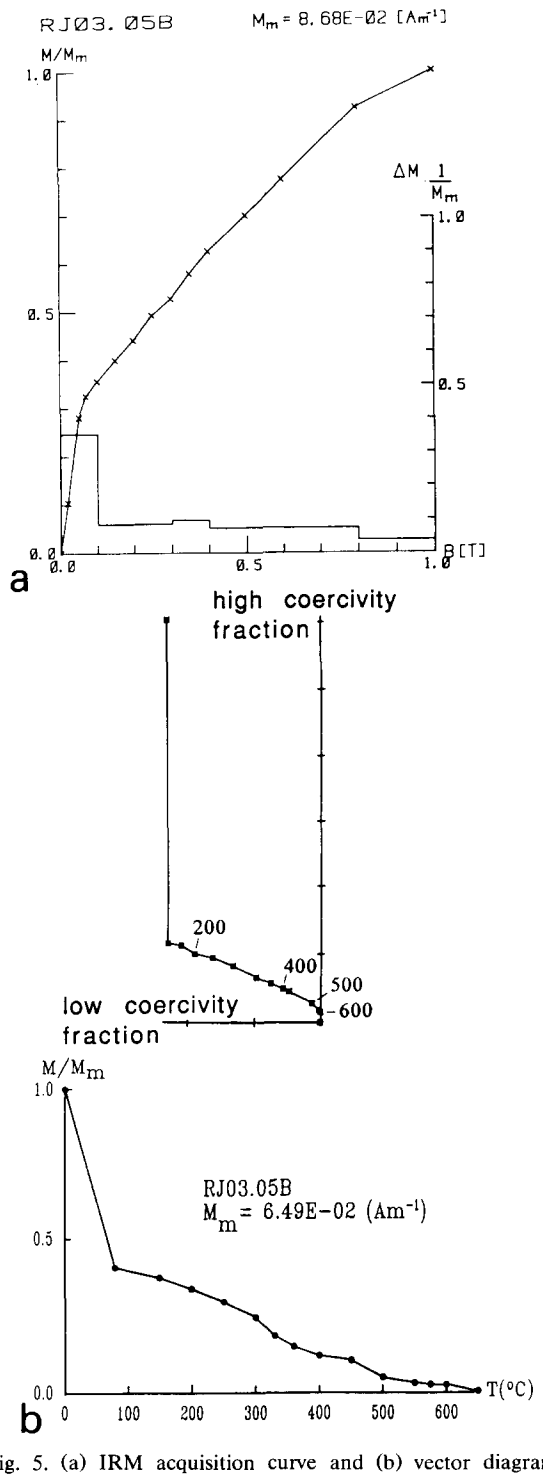


Fig. 5. (a) IRM acquisition curve and (b) vector diagram showing the thermal demagnetization behaviour of a two-component IRM for a sample of grey-white oolitic limestone cored at site RJ03.

350° and 400°C. These minerals, however, probably do not contribute significantly to the remanence characteristics of these limestones.

The NRM of the late Jurassic limestones is composed of two or more components of remanent magnetization (Fig. 6a). Often a lower temperature component, which is unblocked by temperatures below 350°C, yields a present-day magnetic direction. The ChRM is most frequently isolated between 350° and 550°C. Inclinations of this ChRM are on average 43° ± 8°, which is within the range of inclinations expected for the Jurassic of stable Iberia. Declinations, however, deviate from the expected declination (Figs. 2), of 336° calculated from a synthetic polar wander curve for Iberia [45]. The mean declinations for the fourteen Penibetic Jurassic sites is Dec. = 39° ($k = 29$, $\alpha_{95} = 7.8°$). This mean is rotated approximately 63° in a clockwise direction from the expected direction. This result is similar to those obtained by Osete et al. [16], Ogg et al. [18], and Steiner et al. [21] from the Jurassic limestones of the median Subbetic Zone to the north and to the east of the Penibetic realm. The results of their work are shown in Table 2 . (These results have been included In the regional fold test that was performed on the late Jurassic limestones west of Loja (Fig. 7).)

In the regional fold test samples are collected over a large region from beds with different tilts. At each site an individual tilt correction is applied to account for local structure. In the area west of Loja site means from fourteen of our Penibetic sites were combined with five site means obtained by Osete et al. [16], one site mean from Ogg et al. [18] and two from Steiner et al. [21]. The results in Fig. 7 give Dec. = 38° and Inc. = 43° ($N = 22$, $\alpha_{95} = 5.9°$, $k = 29.8$). The significance of the change of the Fisher precision parameters is evaluated by an F-ratio test [46], which shows that the mean direction is significantly improved after applying the tectonic tilt corrections. The ratio of the precision parameters is 4.7, which is greater then the critical value of the F-statistic (2.08) at the 99% confidence level.

4.2. Late Cretaceous marls and limestones

The late Cretaceous Capas Rojas formation throughout the Penibetics is composed of red to grey interbedded marls and marly limestones.

Like the late Jurassic lithologies, these rocks contain stable and well-defined paleomagnetic directions ($\alpha_{95} < 10°$).

The mean intensity of the NRM of 247 samples obtained from these late Cretaceous beds is of the order of 1.52×10^{-3} A/m. The mean direction of NRM is 31° east of the present-day north, indicating a dominance of a rotated remanent component in the uncleaned NRM.

The rock-magnetic characteristics of these rocks are similar to those of the red Jurassic limestones. IRM acquisition studies in conjunc-

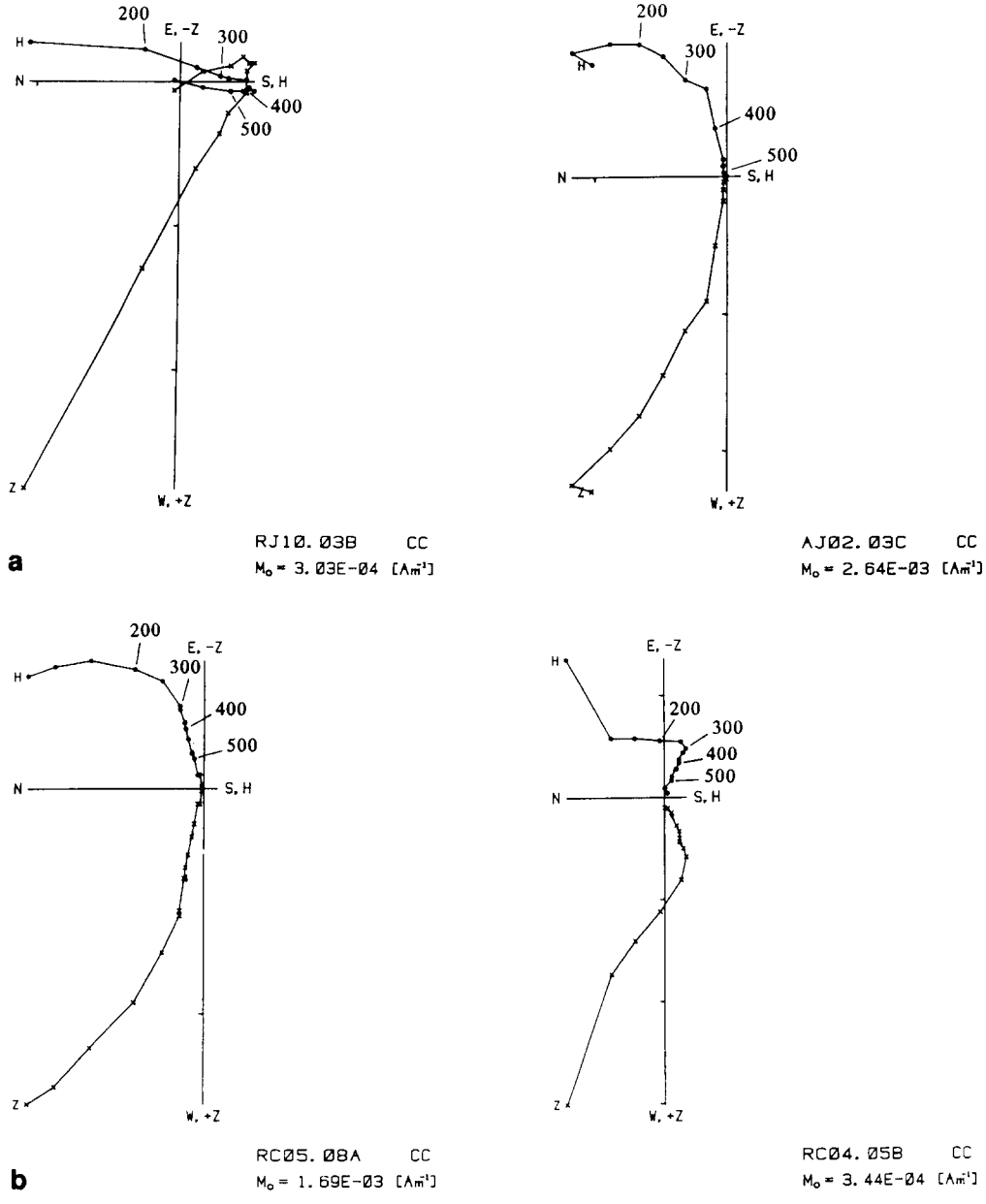


Fig. 6. Typical vector diagrams in geographic coordinates for (a) two late Jurassic carbonate samples and (b) two late Cretaceous carbonate samples undergoing thermal demagnetization. Dots represent vector end-points projected on the horizontal plane; crosses are projections on the vertical N–S plane. Curves show that the NRM is composed of two major components of remanent magnetization.

TABLE 2

Remanent magnetization parameters for previously published sites

Publication	Site	Lat °N	Lon W	N	Dec	Inc	k	α95
Osete et al., 1988	AC	37.34	4.13	14	25.8	43.0	728.7	7.6
	AB	37.34	4.13	6	22.1	17.4	42.2	10.4
	BB	37.40	3.72	11	35.9	32.4	71.7	5.4
	GNV	37.50	3.65	10	32.7	44.7	56.7	6.5
	GNC	37.50	3.65	10	36.3	38.9	15.9	12.5
Ogg et al., 1984	C	37.5	4.3	105	26.3	38.3	66.4	1.7
Steiner et al., 1987	BJ	37.5	4.3	196	53.8	47.0	13.6	2.8
	BC	37.5	4.3	112	59.4	42.9	12.1	4.0

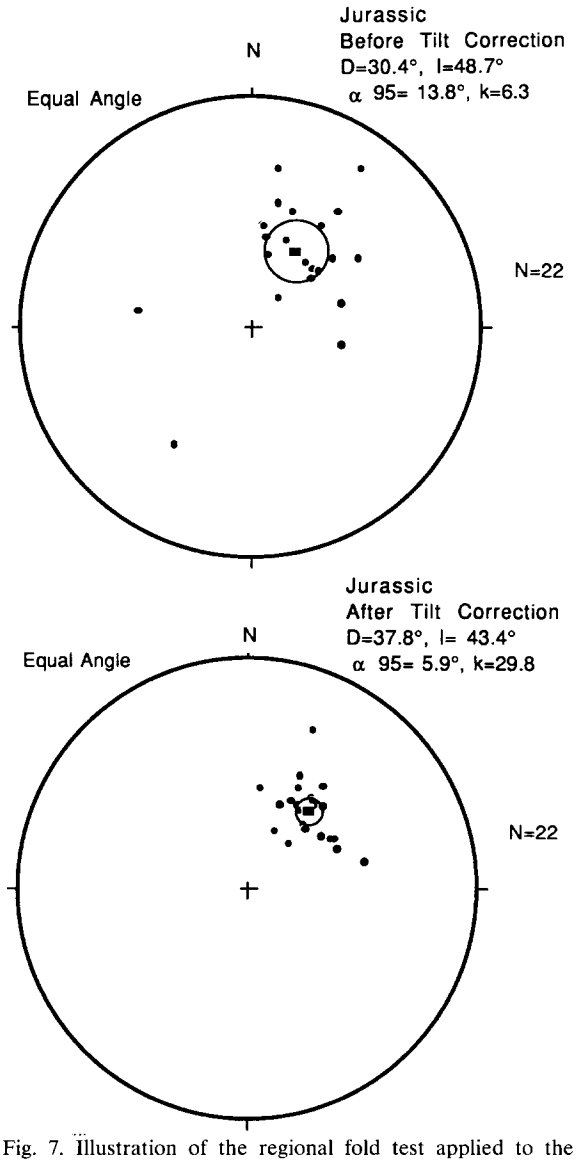


Fig. 7. Illustration of the regional fold test applied to the Jurassic limestones which out crop north of the Campo de Gibraltar flysch basin and west of Granada (excluding the Sierra Gorda). N = number of sites; D = mean declination; I = mean inclination; α_{95} = cone of confidence; k = precision parameter. For this test all site means have been plotted on the lower hemisphere.

tion with thermal demagnetization of an acquired, two-component magnetization reveal the presence of both low and high-coercivity components. These components are successively unblocked at 575°C and 650°–675°C, which suggests that both magnetite and hematite are present in the samples of the late Cretaceous Capas Rojas (Fig. 8a and b).

The NRM is composed of two components of magnetization. A typical Zijderveld diagram for the thermal demagnetization of the late Cretaceous Capas Rojas is presented in Fig. 6b. The low-temperature component of magnetization is usually unblocked by around 300°C and is in the direction of the present-day magnetic field. Sometimes overlap in the unblocking temperature spectra of the lower and higher temperature components leads to a curved segment of the Zijderveld plots in the 300°C range. After removal of this secondary component the ChRM then follows a straight line vector to the origin. This component is unblocked in the temperature range 350°–580°C. These results indicate that the ChRM in the limestones and marls is most probably carried by magnetite. The Zijderveld diagrams suggest that hematite either does not contribute to the ChRM or is in the same direction as the magnetite component.

The mean direction of the ChRM vector for nine Cretaceous sites sampled in the Penibetics is Dec. = 61°, Inc. = 36° (k = 37.3, α_{95} = 9.0°). Inclinations lie within the range of the expected inclination of stable Iberia. The declination, however, deviates from the expected late Cretaceous direction of 358° [45] by an average of 63° in a clockwise sense. An F-test of the Fisher precision parameters for the Cretaceous sites (Fig. 9) shows improvement at the 99% confidence level after applying a tilt correction. In this case the F-ratio was 3.7, which is greater than the critical value of 3.38. However, it must be noted that the statisti-

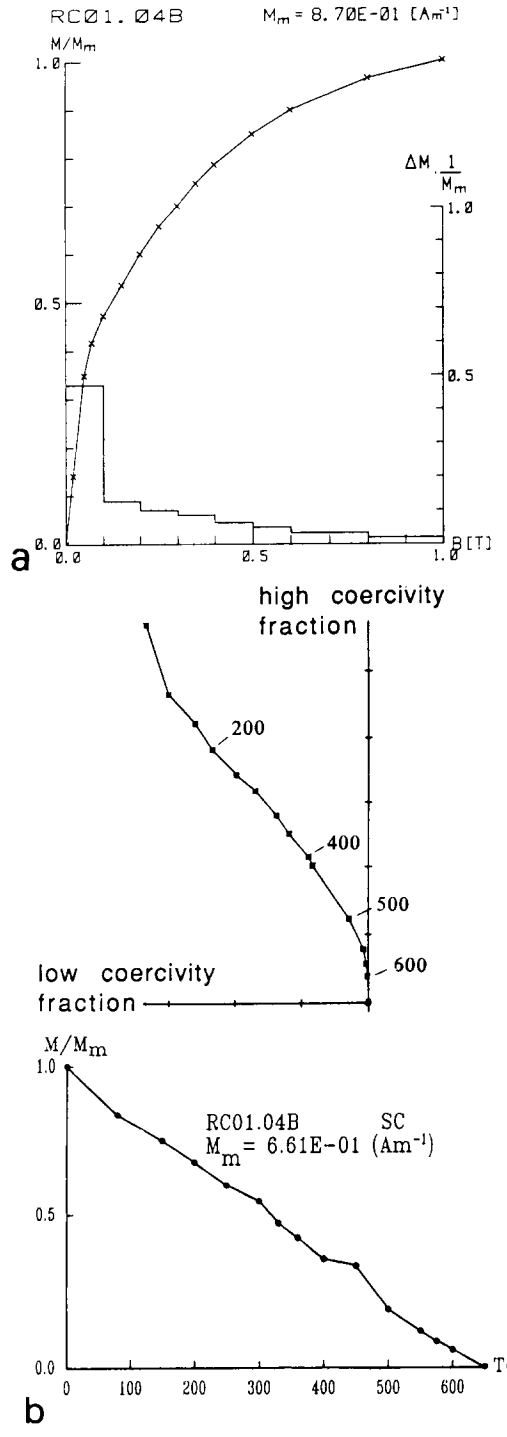


Fig. 8. (a) IRM acquisition curve and (b) vector diagram showing the thermal demagnetization behaviour of a two-component IRM for a typical sample of late Cretaceous marly limestone.

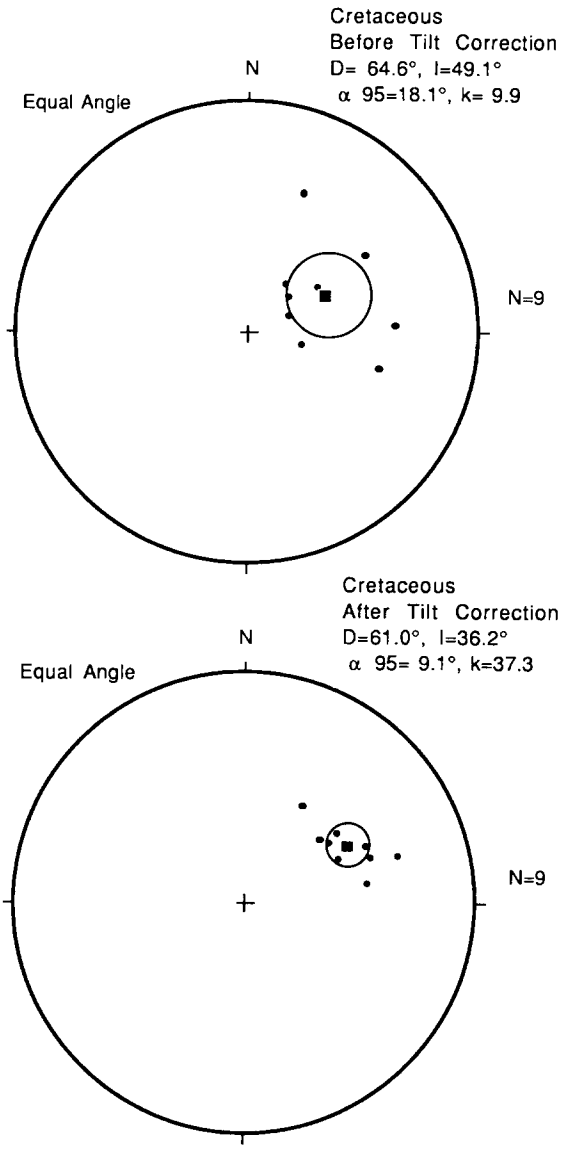


Fig. 9. Illustration of the regional fold test applied to the Cretaceous limestones which outcrop north of the Campo de Gibraltar flysch basin and west of Granada (excluding the Sierra Gorda). Lower hemisphere plot; see Fig. 7 for notation.

cal tests may not be truly applicable because the directions after tilt correction are streaked. This may be an indication of rotation of Iberia during the late Cretaceous deposition of these sediments.

A local fold test was also conducted on this lithology to see if the magnetization predated the mid-Miocene folding event. Figure 10 shows the

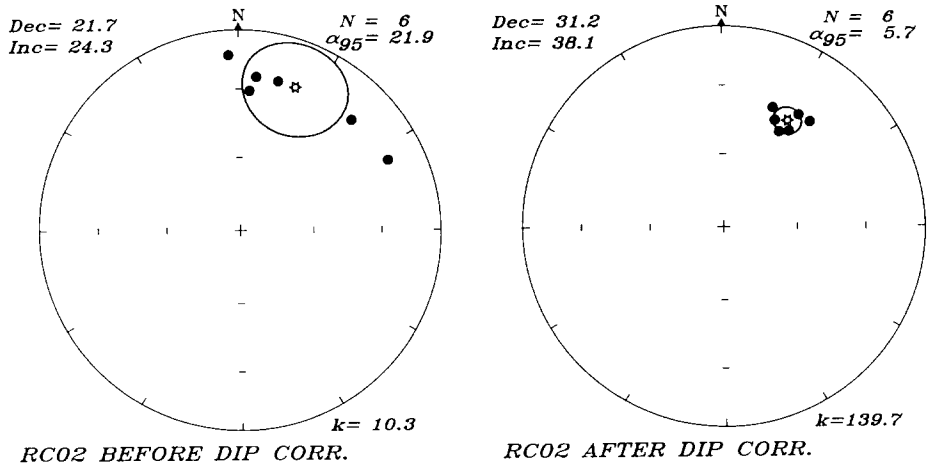


Fig. 10. Illustration of a fold test conducted at site RC02 showing equal area plots of individual ChRM directions, the mean ChRM (star) direction, and the α_{95} both before and after tectonic tilt correction. See Fig. 7 for notation.

results of a positive fold test at one site south of Ronda (RC02).

The Cretaceous rocks are locally highly deformed on the outcrop scale and commonly contain stylolites parallel to bedding. The anisotropy of magnetic susceptibility was measured in these Cretaceous marls and marly limestones to ascertain whether these rocks contained a magnetic fabric which might suggest possible distortion of the NRM directions. The anisotropy measurements show, however, that there is no consistent magnetic fabric in these rocks.

4.3. Data from Mesozoic outliers in the flysch unit

Data obtained from south of the major thrust contact that separates the internal and external units of the Betic Cordillera from the southern flysch deposits come from two areas: Los Canutos and Gibraltar (sites 5 to 34). Both of these areas are isolated blocks surrounded by flysch. Didon [47] suggests that these outcrops may be windows into the substratum of the flysch units.

Los Canutos is a large anticline with a northward plunging axis. Two late Jurassic sites (CJ01, CJ02) indicate a large rotation of this block. This rotation is of the order of 90° in a clockwise sense from the expected direction. Cretaceous sites (CC01, CC02), however, yield clockwise rotations of only 40° from the expected late Cretaceous directions (Fig. 2). This apparent rotation between the Jurassic and Cretaceous cannot be explained from the known kinematic history of Iberia nor can it be explained in the context of the rest of the Betic Cordillera. Unfortunately, the structure of this complex anticline is not known in detail and therefore appropriate tectonic corrections may not have been applied.

5. Discussion: the kinematics of the Iberian Peninsula

5.1. The late Cretaceous rotation of the Iberian Peninsula

The paleomagnetic results obtained in the External Betic Cordillera west of Granada indicate that there is a differential rotation between the Cretaceous sites (Dec. = 61° ± 11°) and the Jurassic sites (Dec. = 38° ± 8°). This differential rotation of 23° between the late Jurassic and late Cretaceous is consistent with the Cretaceous anticlockwise rotation of the Iberian Peninsula with respect to Eurasia and therefore implies that this region was most probably rotated anticlockwise with Iberia between the late Jurassic and the late Cretaceous. The resultant clockwise deviation of the magnetic declination indicates that following the late Cretaceous anticlockwise rotation of Iberia the Mesozoic limestones experienced an additional episode of clockwise rotation. This post-Cretaceous clockwise rotation is about 60°.

The Jurassic and the Cretaceous mean directions were tested to see if they were significantly different at the 95% confidence level [46]. An

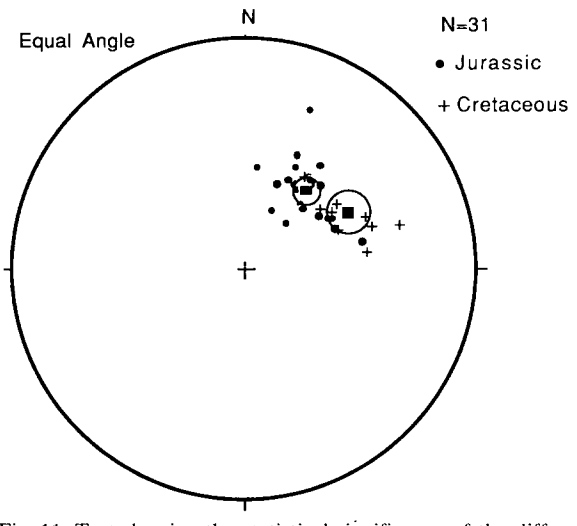


Fig. 11. Test showing the statistical significance of the differential rotation observed between the late Jurassic and the late Cretaceous sites.

F-test on the k parameters of the two populations was conducted to determine whether the populations are statistically the same. In this case the F-parameter is 1.25, which is less than the critical value of 1.89 for (16,42) degrees of freedom. These two populations are comparable. Therefore, seeing as the circles of confidence of the Jurassic and the Cretaceous mean directions do not overlap (Fig. 11) these means are significantly different.

In order to constrain the timing of this late Jurassic to late Cretaceous rotation a local analysis of the stratigraphically continuous section which outcrops at Montejaque was undertaken. The Montejaque section (36°45N′, 5°15′E) is located 3 km north of the town of Montejaque. In this outcrop beds dip approximately 45° to the west to expose an undeformed late Jurassic to late Cretaceous sedimentary section. The biostratigraphy of the Cretaceous part of this section was published by Bourgois [48], and in this section he found that the early Cretaceous and a portion of the late Cretaceous is represented by a hardground or a hiatus. The late Cretaceous begins immediately after this hiatus in the Cenomanian and terminates in the Maastrichtian.

Three sites were drilled at this outcrop. Late Jurassic nodular limestones were sampled along the eastern face of the cliff and two sites of late Cretaceous carbonates were sampled on the western dip slope. Unfortunately, no numerical data were found from the study that was done here approximately thirteen years ago [48], and no detailed cross section was drawn at the locality so it is difficult to correlate precisely the section and the outcrop. However, the ages of the sites based on the proximity of the sites to stratigraphic contacts and on their lithologies can be estimated.

Thermal demagnetization of the two Cretaceous sites yields a ChRM with Dec. = 65.3°, Inc. = 30.3° ($N = 9$, $\alpha_{95} = 5.6°$, $k = 84.1$) for the older site and Dec. = 80.6°, Inc. = 34.1° ($N = 12$, $\alpha_{95} = 3.7°$, $k = 141$) for the younger Cretaceous site. As in the regional study, the values of the precision parameters were compared to show that the k values of the two populations were not significantly different and to establish that their mean directions could be compared. The k ratio, 1.67, was less than the critical value 2.13 of the F-statistic. The two 95% confidence circles do not overlap so the mean declinations for the two populations are significantly different. This suggests that between the Cenomanian (95 Ma) and the Campanian (70 Ma) the block has experienced about 15° of anticlockwise rotation.

In contrast, the late Jurassic mean ChRM has Dec. = 65.7°, Inc. = 44.0° ($N = 8$, $\alpha_{95} = 6.5°$, $k = 74.6$) and is practically identical to the mean Cenomanian direction. These populations are statistically comparable and their α_{95} circles overlap. These values could be considered statistically the same and suggest that no rotation of this block occurred between the late Jurassic and the late Cretaceous. However, the apparent agreement between the mean directions may be coincidence. The confidence limits of the declinations of each mean are about $\pm 10°$, which could allow up to 20° of difference within the resolution of the data.

Earlier paleomagnetic results have implied a total anticlockwise rotation of Iberia of about 35° [6–14]. Unless the observed Cretaceous rotation is a result of local tectonic effects, these data imply that about 15° of the anticlockwise rotation of Iberia occurred between the Cenomanian and the Campanian. This conclusion is consistent with the data obtained from the Bay of Biscay. The magnetic anomalies in the bay indicate that the spreading between France and Iberia began in the late Aptian to Albian and ended in the Cam-

panian. Our results are compatible with those of Galdeano et al. [14] that are based on paleomagnetic results from sediments and volcanics in Portugal; these authors suggested that about 27° of rotation occurred in the early Cretaceous, with about 14° at some later time. The timing of this rotation also agrees with a study by Storetvedt et al. [49] of paleomagnetic and isotopic data from the Portuguese late Cretaceous volcanics and intrusives.

In conclusion, paleomagnetic data from the Penibetics west of Loja indicate that (1) this region was part of the Iberian Peninsula in the late Mesozoic when it rotated anticlockwise away from Europe to open the Bay of Biscay, (2) about 15° of the anticlockwise rotation probably occurred between the Campanian and the Cenomanian, and (3) subsequent to the late Cretaceous these limestones were rotated by at least 60° in a clockwise sense. This major clockwise rotation probably accompanied thrust emplacement during the Alpine orogenic event.

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